Category Theory for Mathematical Physics — Exercise Sheet 5

Fusion categories: particles and how they fuse

Variant: Hints and answers

Conventions as in Lecture 5: $N_{ab}^{c} = \dim \mathrm{Hom}(a \otimes b, c)$, the fusion matrix N_a has entries $(N_a)_{cb} = N_{ab}^{c}$, and d_a is the quantum dimension of the canonical spherical structure (Theorem 5.5). Permutations compose right to left, $(gh)(i) = g(h(i))$, and S_3 acts on $\mathbb{C}^3$ by permuting coordinates. The level is an integer $k \geq 1$, the label j of $\mathrm{SU}(2)_k$ is a spin, $0 \leq j \leq k/2$, $\theta = \pi/(k+2)$, and $\varphi = (1+\sqrt{5})/2$. A splitting vertex carries its labels, ψ_c^{ab}; the bare ψ is Ising's fermion. The f_n of Exercise 4 are Fibonacci numbers.

Exercise 1 (Two fusion rings from S_3). (i) In Vec_{S_3} (trivial ω) show that $N_g N_h = N_{gh}$ for all $g, h \in S_3$, deduce associativity of the fusion ring, and show that $N_{(12)}$ and $N_{(23)}$ do not commute.

(ii) Draw the bending picture of Theorem 5.3 that turns $f\colon a \otimes b \to c$ into a morphism $b \otimes c^* \to a^*$. Then check $N_{ab}^{c} = N_{bc^*}^{a^*}$ directly: in Vec_{S_3} for all g, h, k, and in $\mathrm{Rep}(S_3)$ from the fusion rules of Example 5.2.

Hint. (i) N_g is a permutation matrix: $(N_g)_{cb} = 1$ exactly when $c = gb$. (ii) In $\mathrm{Rep}(S_3)$ every simple object is self-dual, so the identity says that N_{ab}^{c} is unchanged by a cyclic shift of (a, b, c).

Answer. (i) $N_g N_h$ and N_{gh} both send the basis vector b to ghb; $N_{(12)}N_{(23)} = N_{(123)} \neq N_{(132)} = N_{(23)}N_{(12)}$. (ii) $gh = k$ exactly when $hk^{-1} = g^{-1}$; in $\mathrm{Rep}(S_3)$ the non-zero N_{ab}^{c} are those with $\{a, b, c\}$ one of $\{1, 1, 1\}$, $\{1, \mathrm{sgn}, \mathrm{sgn}\}$, $\{1, \mathrm{std}, \mathrm{std}\}$, $\{\mathrm{sgn}, \mathrm{std}, \mathrm{std}\}$, $\{\mathrm{std}, \mathrm{std}, \mathrm{std}\}$ (as multisets), each 1.

Exercise 2 (Quantum dimensions of $\mathrm{SU}(2)_k$). Let $d_j = \sin\big((2j+1)\theta\big)/\sin\theta$ for $j = -\frac{1}{2}, 0, \frac{1}{2}, \ldots, \frac{k+1}{2}$.

(i) Show that $d_{-1/2} = d_{(k+1)/2} = 0$ and that

$$d_{1/2}\, d_j = d_{j-1/2} + d_{j+1/2} \qquad \big(0 \leq j \leq \tfrac{k}{2}\big),$$

and explain why this matches the truncated rule for $\frac{1}{2} \otimes j$, including at $j = \frac{k}{2}$.

(ii) Show that every $[j]$ in the fusion ring is a polynomial in $[\frac{1}{2}]$ with integer coefficients. Show that an additive map f from the fusion ring to $\mathbb{R}$ with $f([0]) = 1$ and $f([\frac{1}{2}]z) = f([\frac{1}{2}])f(z)$ for every z is multiplicative. Conclude with Theorem 5.4 that $d_j = \mathrm{FPdim}(j)$.

(iii) List the d_j for $k = 2, 3, 4$.

Hint. (i) $2\cos\theta\, \sin x = \sin(x+\theta) + \sin(x-\theta)$. (ii) Read the rule for $\frac{1}{2} \otimes j$ as a formula for $[j + \frac{1}{2}]$; for multiplicativity, first show $f([\frac{1}{2}]^n z) = f([\frac{1}{2}])^n f(z)$.

Answer. (i) $d_{1/2} = 2\cos\theta$, and the identity is $2\cos\theta \sin\big((2j+1)\theta\big) = \sin\big((2j+2)\theta\big) + \sin(2j\theta)$; $d_{(k+1)/2} = \sin\pi/\sin\theta = 0$ removes the truncated channel. (ii) $[j+\frac{1}{2}] = [\frac{1}{2}][j] - [j-\frac{1}{2}]$; d is positive and multiplicative, hence FPdim. (iii) $k=2$: $1, \sqrt{2}, 1$; $k=3$: $1, \varphi, \varphi, 1$; $k=4$: $1, \sqrt{3}, 2, \sqrt{3}, 1$.

Exercise 3 (Vertices for std ⊗ std). Realise std on $\{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$ with the orthonormal basis $v_1 = (e_1 - e_2)/\sqrt{2}$, $v_2 = (e_1 + e_2 - 2e_3)/\sqrt{6}$, and write $v_{ab} = v_a \otimes v_b$. Let

$$w_{\mathbf{1}}(1) = \tfrac{1}{\sqrt{2}}(v_{11} + v_{22}), \qquad w_{\mathrm{sgn}}(1) = \tfrac{1}{\sqrt{2}}(v_{12} - v_{21}),$$
$$w_{\mathrm{std}}(v_1) = \tfrac{1}{\sqrt{2}}(v_{12} + v_{21}), \qquad w_{\mathrm{std}}(v_2) = \tfrac{1}{\sqrt{2}}(v_{11} - v_{22}).$$

(i) Write the matrices of (12) and (23) on std, and show that each w_c is an isometric intertwiner $c \to \mathrm{std} \otimes \mathrm{std}$.

(ii) With the splitting vertices $\psi_c^{\mathrm{std}\,\mathrm{std}} = (4/d_c)^{1/4}\, w_c$, check the bubble Equation (5.1) for all c, c', and the theta graphs Equation (5.4).

(iii) Draw the completeness relation Equation (5.3) for std ⊗ std, and check it as an identity of 4×4 matrices.

Hint. (i) (12) and (23) generate S_3. (ii) Start from the products $w_c^\dagger w_{c'}$. (iii) What kind of map is $w_c w_c^\dagger$?

Answer. (i) $(12) \mapsto \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, $(23) \mapsto \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}$; the images are orthonormal. (ii) $(\psi_c^{\mathrm{std}\,\mathrm{std}})^\dagger \psi_{c'}^{\mathrm{std}\,\mathrm{std}} = \delta_{cc'}\,(2/\sqrt{d_c})\,\mathrm{id}_c$; theta graphs 2, 2, $2\sqrt{2}$. (iii) The three projections sum to I_4.

Exercise 4 (How fusion spaces grow). (i) Draw the fusion trees (Theorem 5.6) with four leaves τ and root τ in the Fibonacci category.

(ii) Let g_n be the vector $(\dim V_1^{\tau\cdots\tau}, \dim V_\tau^{\tau\cdots\tau})$ for $n \geq 1$ leaves τ. Show that $g_n = N_\tau^{n-1} g_1$ with $g_1 = (0, 1)$, and hence $g_n = (f_{n-1}, f_n)$ with $f_0 = 0$, $f_1 = 1$, $f_{n+1} = f_n + f_{n-1}$. Prove Binet's formula $f_n = \big(\varphi^n - (-1/\varphi)^n\big)/\sqrt{5}$, and deduce that $f_{n+1}/f_n \to \varphi = d_\tau$. How many states do 20 τ's with total charge τ have?

(iii) In the Ising category, show that for $m \geq 1$, $2m$ copies of σ have 2^{m-1} states with total charge 1 and 2^{m-1} with total charge ψ.

Hint. (i) Label the two inner edges; a 1 cannot follow a 1. (ii) The last vertex of a tree fuses the previous total charge with one more τ; φ and $-1/\varphi$ are the roots of $x^2 = x + 1$. (iii) Look at N_σ^2 on the span of 1 and ψ.

Answer. (i) Three trees, with inner labels $(x_1, x_2) = (1, \tau)$, $(\tau, 1)$, (τ, τ). (ii) $g_{n+1} = N_\tau g_n$; $f_{20} = 6765$ states. (iii) N_σ^2 maps $(u, 0, v)$ to $(u + v, 0, u + v)$.

Exercise 5 (D_8 and Q_8). Let G be D_8 or Q_8, given by the 2×2 matrices of Sheet 4, acting on $V = \mathbb{C}^2$. You may use, from *Symmetries* (cited in Lecture 5), that irreducible characters are orthonormal for $\langle \chi, \chi' \rangle = \frac{1}{|G|} \sum_g \chi(g)\overline{\chi'(g)}$, that the irreducible representations have $\sum \dim^2 = |G|$, and that $N_{ab}^c = \langle \chi_a \chi_b, \chi_c \rangle$; and that in both groups $\{\pm 1\}$ is a normal subgroup with quotient $\mathbb{Z}_2 \times \mathbb{Z}_2$.

(i) Show that G has exactly five irreducible representations: four of dimension 1, pulled back from $\mathbb{Z}_2 \times \mathbb{Z}_2$ (call them $\mathbf{1}, \chi_1, \chi_2, \chi_3$, with $\chi_1\chi_2 = \chi_3$), and V.

(ii) Compute all fusion rules of $\mathrm{Rep}(G)$, and show that $\mathrm{Rep}(D_8)$ and $\mathrm{Rep}(Q_8)$ have the same fusion ring.

(iii) Give the FPdim's and the global dimension.

Hint. (i) Sheet 4 gives χ_V: 2 at 1, -2 at -1, 0 elsewhere. (ii) χ_V vanishes off $\{\pm 1\}$, and every χ_i is 1 on $\{\pm 1\}$.

Answer. (i) $\langle \chi_V, \chi_V \rangle = (4{+}4)/8 = 1$, and $4 \cdot 1^2 + 2^2 = 8$. (ii) The χ's fuse as $\mathbb{Z}_2 \times \mathbb{Z}_2$, $\chi_i \otimes V \cong V$, $V \otimes V \cong \mathbf{1} \oplus \chi_1 \oplus \chi_2 \oplus \chi_3$, for both groups. (iii) FPdim = 1, 1, 1, 1, 2; global dimension 8.

Exercise 6 ⋆. *Global dimensions of the four families.* Recall $\dim \mathcal{C} = \sum_a d_a^2$ with the canonical $d_a = \mathrm{FPdim}(a)$.

(i) Show that $\dim \mathrm{Vec}_G = \dim \mathrm{Rep}(G) = |G|$, $\dim \mathrm{Fib} = 2 + \varphi$ and $\dim \mathrm{Ising} = 4$.

(ii) Show that $\sum_{m=1}^{k+1} \sin^2(m\theta) = \frac{k+2}{2}$, and deduce

$$\dim \mathrm{SU}(2)_k = \frac{k+2}{2 \sin^2 \theta}.$$

Evaluate it for $k = 2, 3, 4$.

(iii) Show that for every k the object $\frac{k}{2}$ is invertible, with $\frac{k}{2} \otimes j \cong \frac{k}{2} - j$, and deduce $d_{k/2-j} = d_j$.

(iv) Let k be odd. Show that the integer spins span a fusion subring, that every simple object is $y \otimes x$ for exactly one $y \in \{0, \frac{k}{2}\}$ and one integer spin x, and that

$$\dim \mathrm{SU}(2)_k = 2 \sum_{j \in \mathbb{Z}} d_j^2.$$

What is the integer-spin subring for $k = 3$? Why does the argument fail for even k? Show that the identity holds for even k all the same, with an alternating version of the sum in (ii).

Hint. (i) For $\mathrm{Rep}(G)$, FPdim is the dimension. (ii) Write $\sin^2$ through $\cos 2x$. (iii) Both bounds of the truncated rule. (iv) For even k, consider $\sum_m (-1)^m \sin^2(m\theta)$.

Answer. (i) $d_g = 1$ for $|G|$ objects; $\sum \dim^2 = |G|$; $1 + \varphi^2 = 2 + \varphi$; $1 + 2 + 1 = 4$. (ii) $\dim \mathrm{SU}(2)_k = \frac{1}{\sin^2 \theta} \sum_{m=1}^{k+1} \sin^2(m\theta)$; 4, $2(2 + \varphi)$, 12 for $k = 2, 3, 4$. (iii) The rule leaves the single channel $\frac{k}{2} - j$; $d_{k/2} = 1$ and multiplicativity give $d_{k/2-j} = d_j$. (iv) $j \mapsto \frac{k}{2} - j$ swaps integer and half-integer spins; for $k = 3$ the subring is Fibonacci's, on $0, 1$, and $\dim \mathrm{SU}(2)_3 = 2(2 + \varphi)$; for even k, $\frac{k}{2}$ is itself an integer spin, but $\sum_{m=1}^{k+1} (-1)^m \sin^2(m\theta) = 0$ still splits the sum in half.